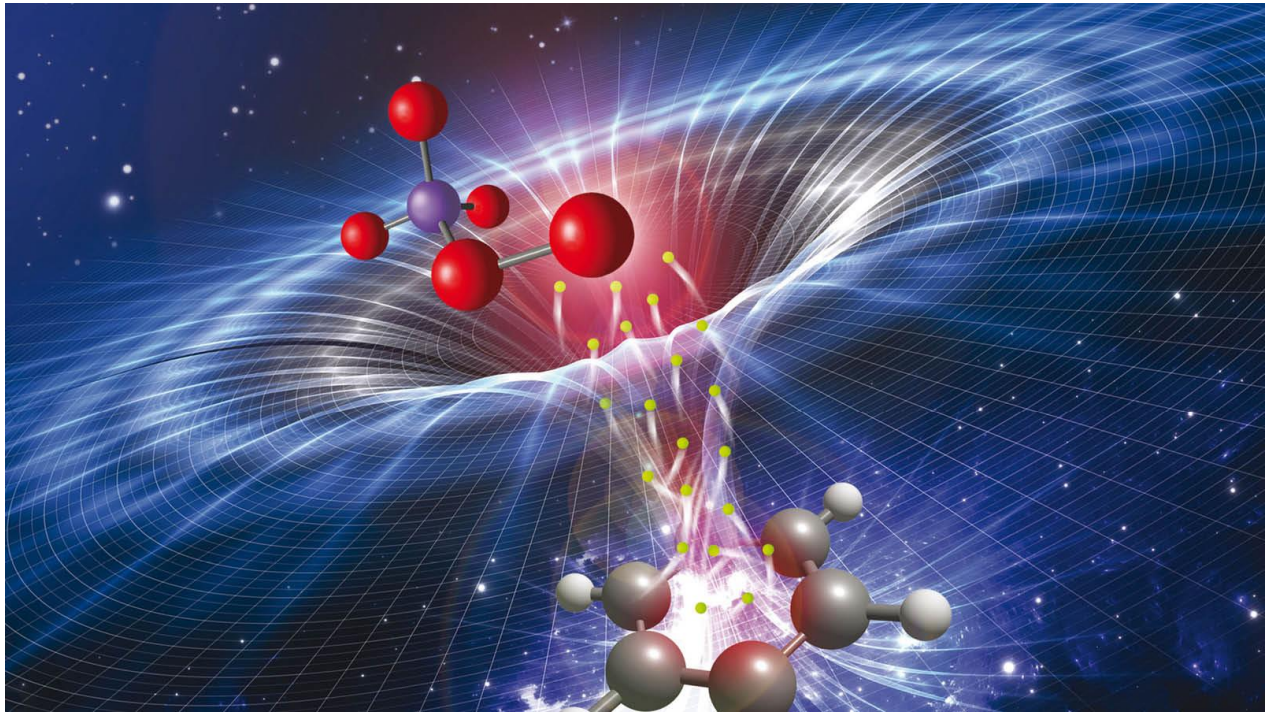
General Chemistry (C102)



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State of matter
Lecture 3

Real Gas

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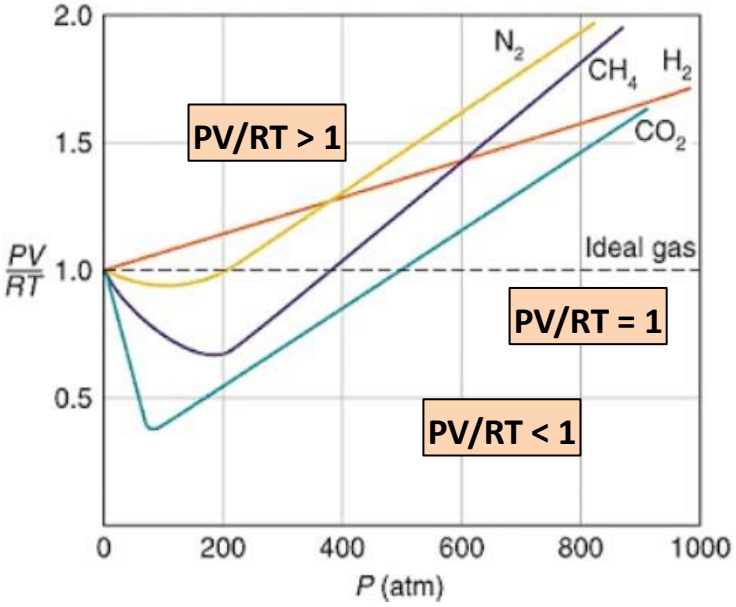
Deviations from Ideal Behaviour

- An **ideal gas** is one which obeys the gas laws or the gas equation $PV = RT$ at all pressures and temperatures. However, no gas is ideal. Almost all gases show deviations from the ideal behaviour. Thus, the gases H_2, N_2 and CO_2 which fail to obey the ideal-gas equation are termed **nonideal** or **real gases.**



✓ **Repulsive force are dominant**

$PV_{real} > PV_{ideal}$

✓ **The attractive forces are dominant**

$PV_{real} < PV_{ideal}$

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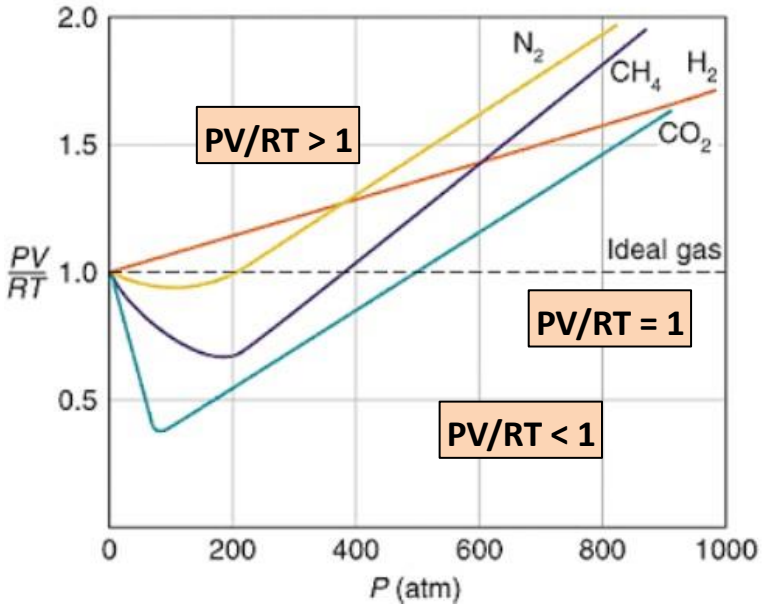
Deviations from Ideal Behaviour

- The following figure shows the compressibility factor, Z (PV/RT), plotted against pressure for H_2, N_2 and CO_2 at a constant temperature.
- **At very low pressure,** for all these gases *Z* is approximately equal to one. This indicates that at low pressures (up to 10 atm), real gases exhibit nearly ideal behaviour.



✓ **Repulsive force are dominant**

$PV_{real} > PV_{ideal}$

✓ **The attractive forces are dominant**

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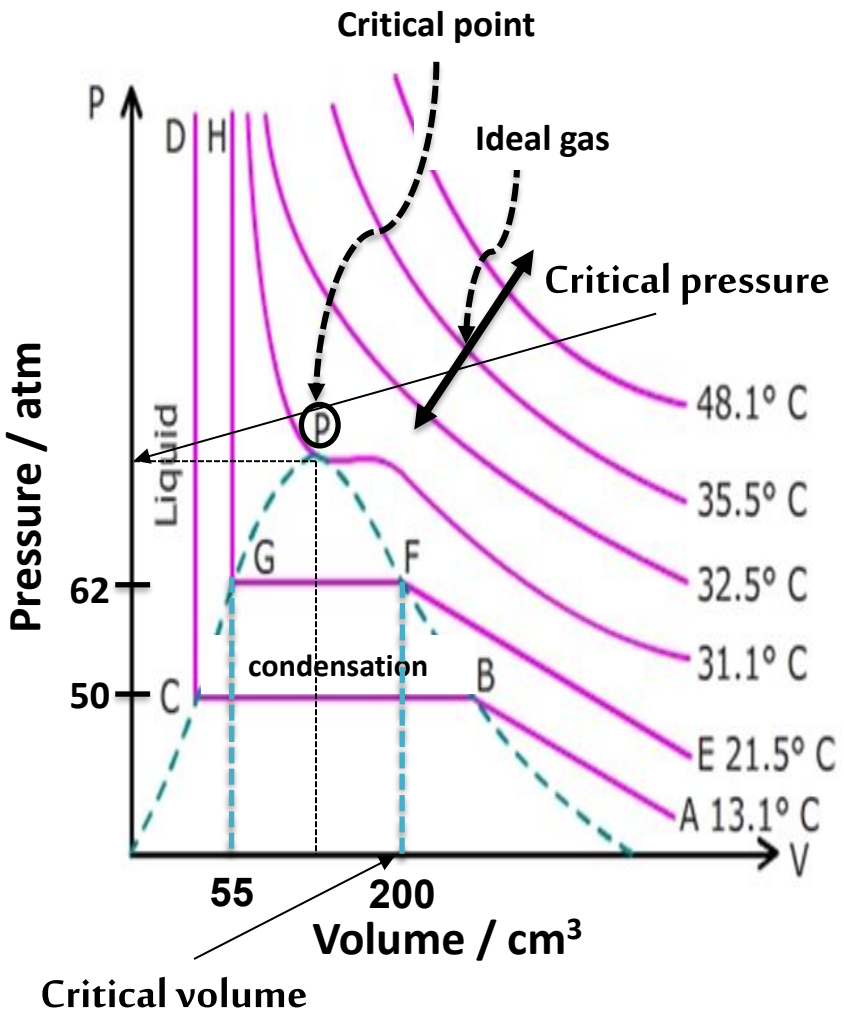
Andrews experiment on carbon dioxide

The curves obtained by Andrews experiment are shown in the Figure. We observe that there are two types of isothermals:

(a) **Above 31.1°C**, the isothermals (*at constant Temperature*) are almost rectangular hyperbola resembling the behaviour of ideal gas.

(b) **Below 31.1°C**, the isothermals are not rectangular hyperbola.



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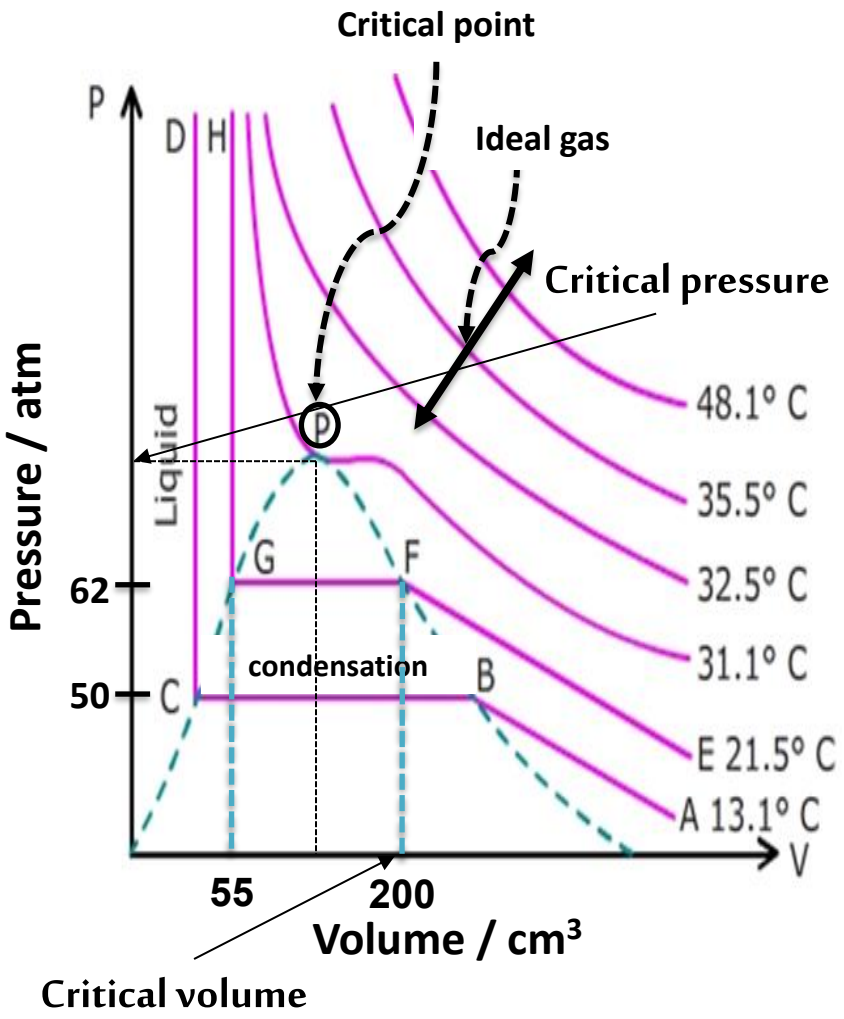
Andrews experiment on carbon dioxide

- At 13.1°C, the portion **AB** of this curve shows that with the increase of pressure, volume decreases up to the point B showing gaseous behaviour of CO_2 (Boyle's law is obeyed).
- From **B** to **C**, there is change of state of CO_2 from gaseous to liquid (**condensation**) and the curve becomes parallel to the volume axis. This indicates that volume decreases without appreciable increase in pressure. After the point **C**, the curve is very steep indicating that after **C**, the substance becomes highly incompressible. At **C**, the gas gets liquefied completely. The portion **CD** represents the liquid phase of CO_2.



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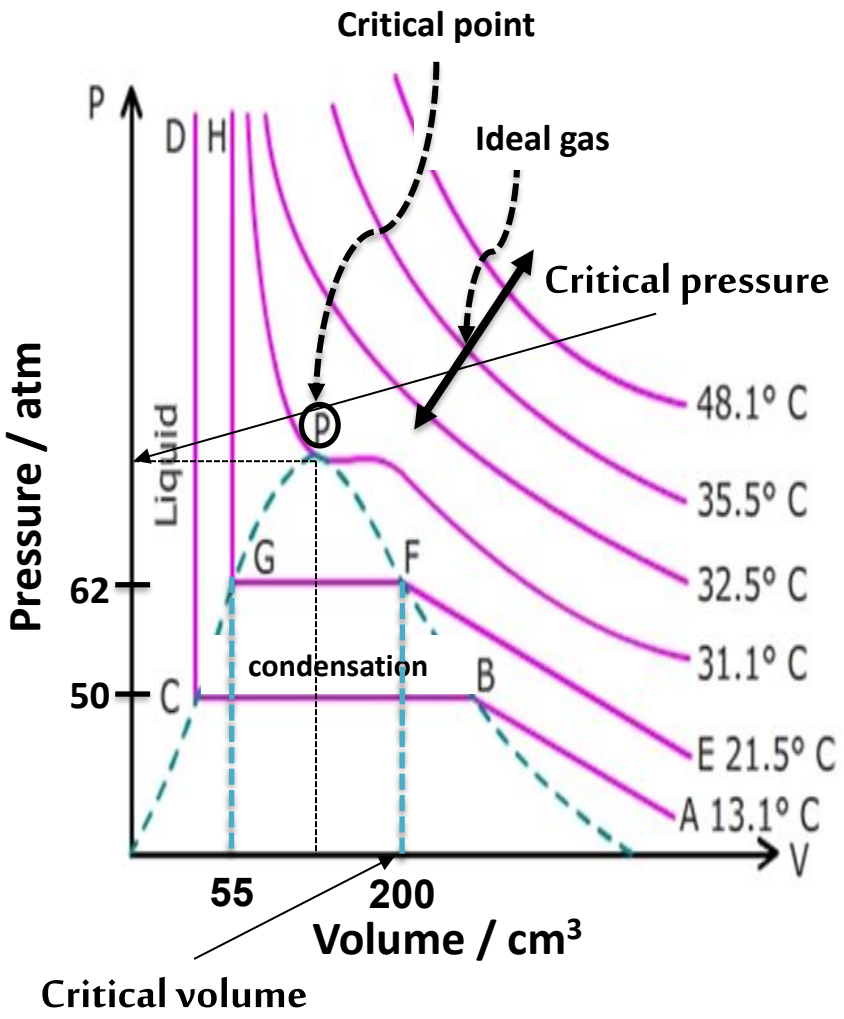
Critical constants/Definitions

1. Critical Temperature: (T_C)

The temperature at which it just becomes possible to liquefy a gas under compression is known as the 'critical temperature'. Above this temperature the gas cannot be liquefied even when large pressure is applied.

2. Critical pressure: (P_C)

It is the pressure necessary to liquefy a gas at critical temperature.



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Critical constants

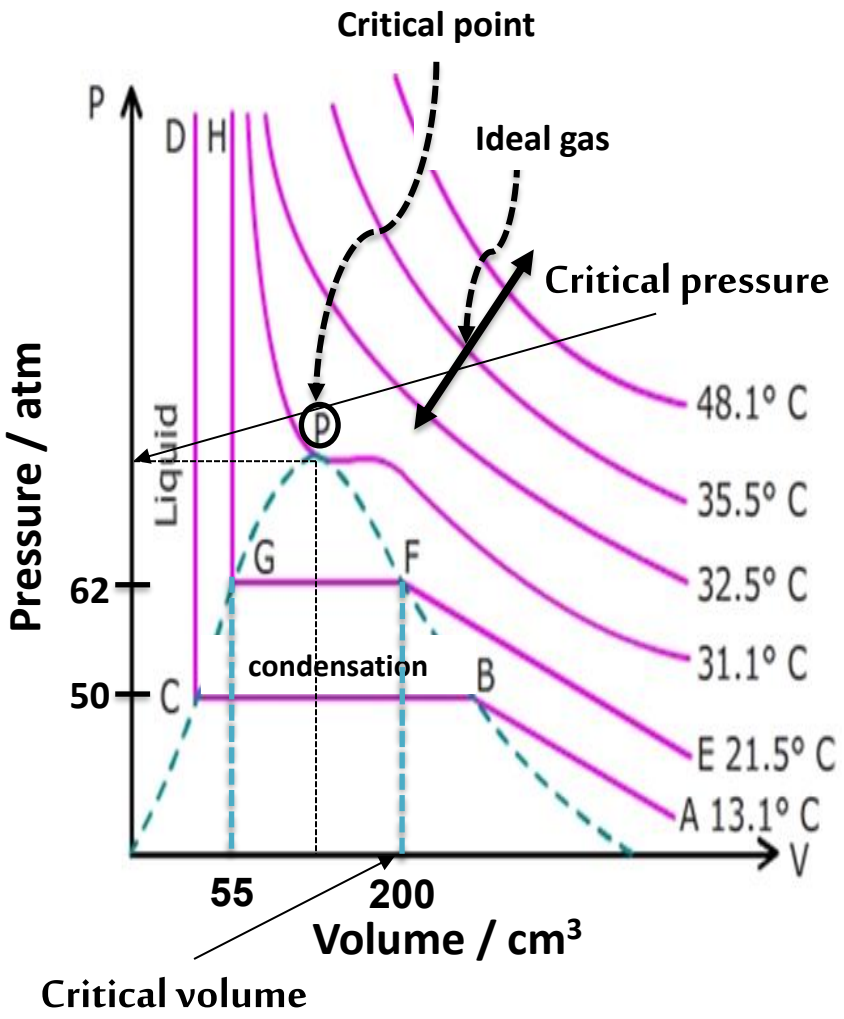
3. Critical Volume: (V_C)

It is the volume which unit mass (or one mole of a gas) of a gas occupies at the critical temperature and pressure.

4. Critical Point:

It is the point on the isothermal for the critical temperature at which the change of state from gaseous to the liquid takes place under constant values of P_C and V_C.

T_C, P_C and V_C are collectively known as 'critical constants'.



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Real Gases

- Van Der Waals (1873) attributed the deviations of real gases from ideal behaviour to two erroneous postulates of the kinetic theory. These are:

  1. **The molecules in a gas are point masses and possess no volume.**

  2. **There are no intermolecular forces in a gas.**

- Therefore, the ideal gas equation $PV = nRT$ derived from kinetic theory could not hold for real gases. Van Der Waals pointed out that both the **pressure (*P*)** and **volume (*V*)** factors in the ideal gas equation needed correction in order to make it applicable to real gases.

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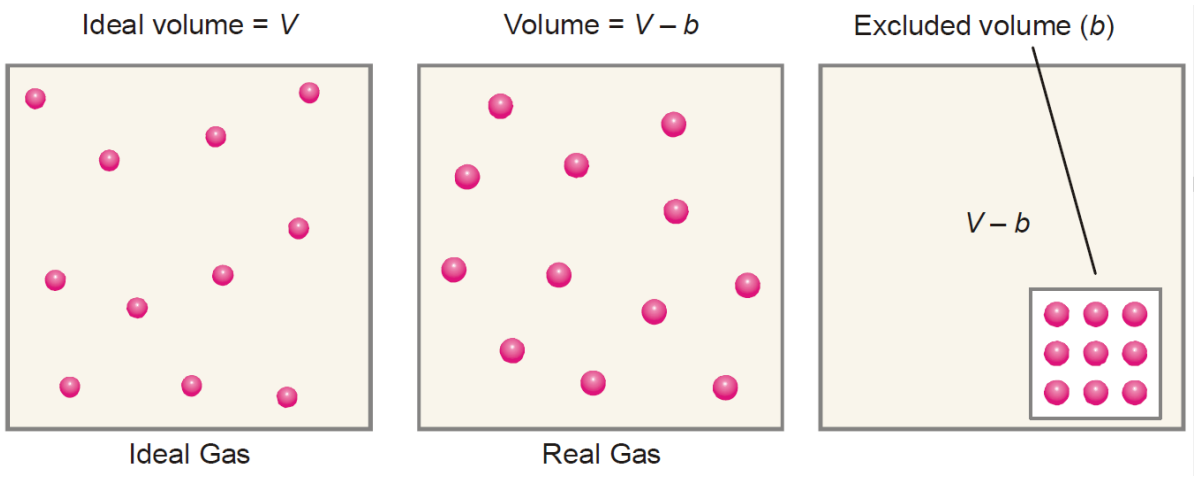
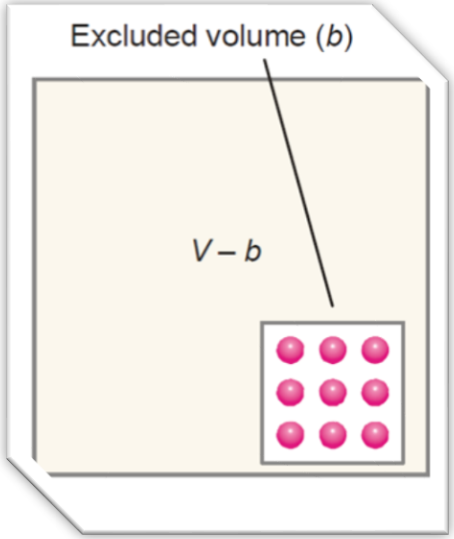
Volume Correction

- The volume of a gas is the free space in the container in which molecules move about. **Volume (V) of an ideal gas is the same as the volume of the container.** The dot molecules of ideal gas have zero volume and the entire space in the container is available for their movement. However, Van Der Waals assumed that molecules of a real gas are rigid spherical particles which possess a definite volume.





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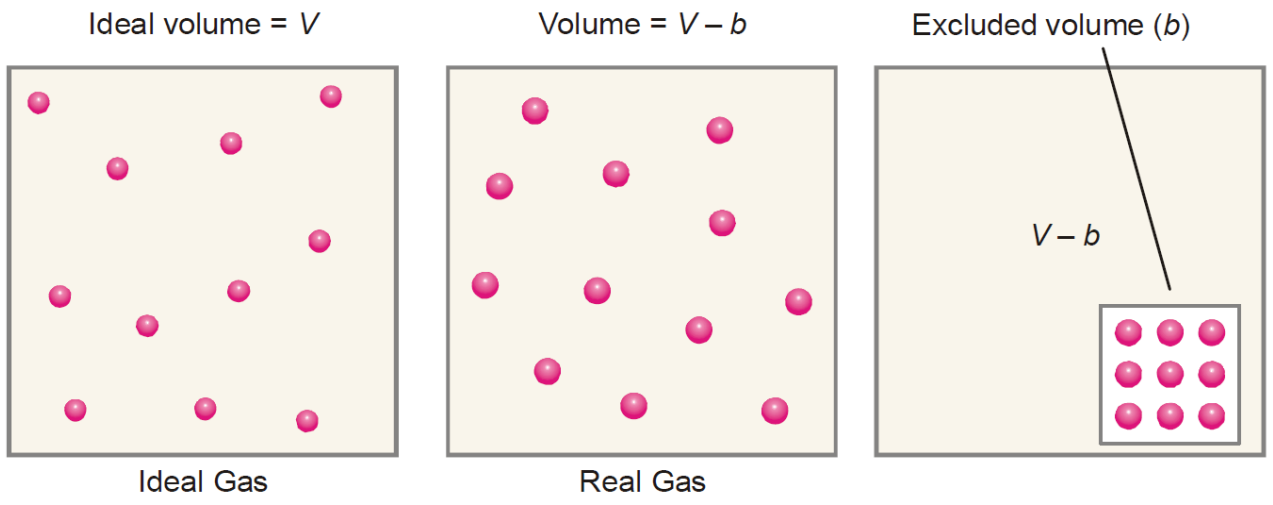
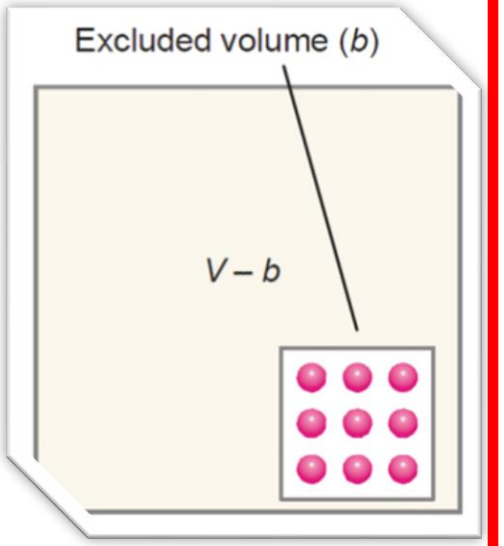
Volume Correction

- **The volume of a real gas is, therefore, ideal volume minus the volume occupied by gas molecules.** If b is the effective volume of molecules per mole of the gas, the volume in the ideal gas equation is corrected as: $(V - b)$ For n moles of the gas, the corrected volume is: $(V - nb)$ Where, b is termed the **excluded volume** which is constant and characteristic for each gas.



Corrected volume

$V' = V - nb$



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Volume Correction

✓ A molecule in the interior of a gas is attracted by other molecules on all sides. These attractive forces cancel out. But a molecule about to strike the wall of the vessel is attracted by molecules on one side only. Hence, it experiences an inward pull (Figure a). Therefore, it strikes the wall with reduced velocity and the actual pressure of the gas, P, will be less than the ideal pressure.

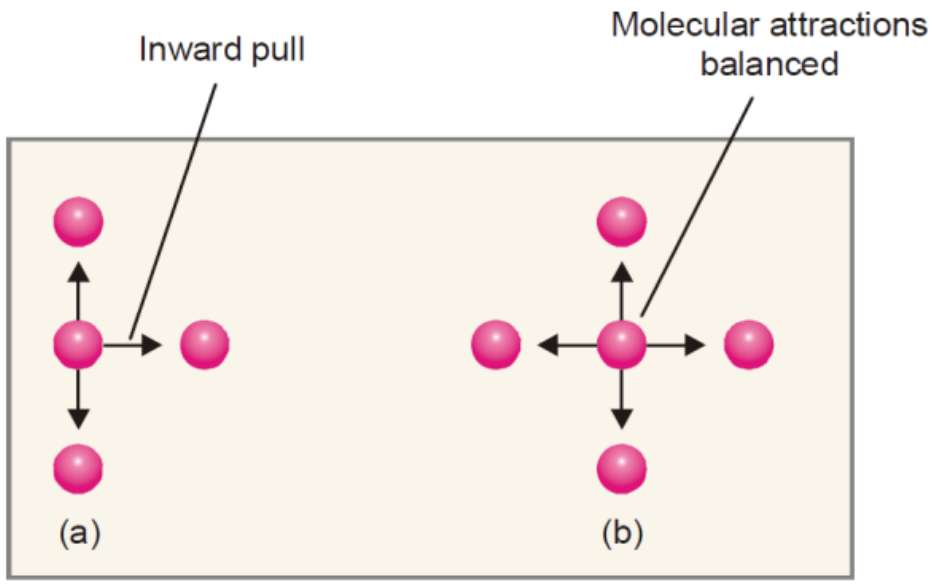

Figure:

a) **A molecule about to strike the wall has a net inward pull**

b) **A molecule in the interior of gas has balanced attractions.**

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Pressure Correction

- If the actual pressure P, is less than P_{ideal} by a quantity p, we have

$$P = P_{ideal} - p \qquad \text{Or,} \quad P_{ideal} = P + p$$

➢ p is determined by the force of attraction between molecules (A) striking the wall of container and the molecules (B) pulling them inward.

➢ The net force of attraction is, therefore, proportional to the concentration of (A) type molecules and also of (B) type of molecules. That is,

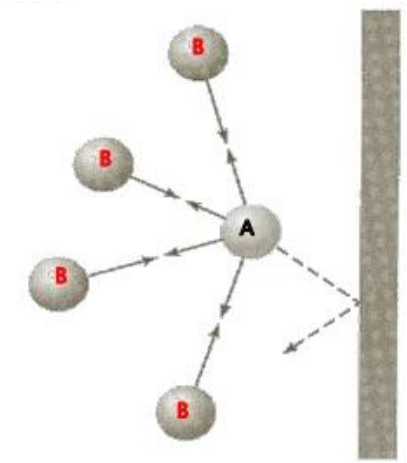
$$p \propto C_A \times C_B$$

$$Or, p \propto \frac{n}{V} \times \frac{n}{V}$$

$$Or, p = \frac{an^2}{V^2}$$

- **Where, *n* is total number of gas molecules in volume *V* and '*a*' is proportionality constant characteristic of the gas.**
- **Thus, the pressure *P* in the ideal gas equation is corrected as**

$$P_{ideal} = P + a\left(\frac{n}{V}\right)^2$$



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Van der Waal's equation

- Substituting the values of corrected pressure and volume in the ideal gas equation, PV = nRT, we get.

$$[P + a\left(\frac{n}{V}\right)^2](V - nb) = nRT$$

- This is known as **Van Der Waals** equation for n moles of a gas.
- "**a**" and "**b**" are **determined by experiment.**
- "**a**" and "**b**" are **different for each gas.**
- **Bigger molecules** have **larger "b"**.
- "**a**" depends on both **size and polarity.**

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